CAMBRIDGE
International Education

Cambridge O Level

ADDITIONAL MATHEMATICS **4037/23**

Paper 2 **October/November 2024**

MARK SCHEME

Maximum Mark: 80

Published

This mark scheme is published as an aid to teachers and candidates, to indicate the requirements of the examination. It shows the basis on which Examiners were instructed to award marks. It does not indicate the details of the discussions that took place at an Examiners' meeting before marking began, which would have considered the acceptability of alternative answers.

Mark schemes should be read in conjunction with the question paper and the Principal Examiner Report for Teachers.

Cambridge International will not enter into discussions about these mark schemes.

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[Turn over

Generic Marking Principles

These general marking principles must be applied by all examiners when marking candidate answers. They should be applied alongside the specific content of the mark scheme or generic level descriptions for a question. Each question paper and mark scheme will also comply with these marking principles.

GENERIC MARKING PRINCIPLE 1:

Marks must be awarded in line with:

- the specific content of the mark scheme or the generic level descriptors for the question
- the specific skills defined in the mark scheme or in the generic level descriptors for the question
- the standard of response required by a candidate as exemplified by the standardisation scripts.

GENERIC MARKING PRINCIPLE 2:

Marks awarded are always **whole marks** (not half marks, or other fractions).

GENERIC MARKING PRINCIPLE 3:

Marks must be awarded **positively**:

- marks are awarded for correct/valid answers, as defined in the mark scheme. However, credit is given for valid answers which go beyond the scope of the syllabus and mark scheme, referring to your Team Leader as appropriate
- marks are awarded when candidates clearly demonstrate what they know and can do
- marks are not deducted for errors
- marks are not deducted for omissions
- answers should only be judged on the quality of spelling, punctuation and grammar when these features are specifically assessed by the question as indicated by the mark scheme. The meaning, however, should be unambiguous.

GENERIC MARKING PRINCIPLE 4:

Rules must be applied consistently, e.g. in situations where candidates have not followed instructions or in the application of generic level descriptors.

GENERIC MARKING PRINCIPLE 5:

Marks should be awarded using the full range of marks defined in the mark scheme for the question (however; the use of the full mark range may be limited according to the quality of the candidate responses seen).

GENERIC MARKING PRINCIPLE 6:

Marks awarded are based solely on the requirements as defined in the mark scheme. Marks should not be awarded with grade thresholds or grade descriptors in mind.

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Mathematics-Specific Marking Principles

1 Unless a particular method has been specified in the question, full marks may be awarded for any correct method. However, if a calculation is required then no marks will be awarded for a scale drawing.

2 Unless specified in the question, non-integer answers may be given as fractions, decimals or in standard form. Ignore superfluous zeros, provided that the degree of accuracy is not affected.

3 Allow alternative conventions for notation if used consistently throughout the paper, e.g. commas being used as decimal points.

4 Unless otherwise indicated, marks once gained cannot subsequently be lost, e.g. wrong working following a correct form of answer is ignored (isw).

5 Where a candidate has misread a number or sign in the question and used that value consistently throughout, provided that number does not alter the difficulty or the method required, award all marks earned and deduct just 1 A or B mark for the misread.

6 Recovery within working is allowed, e.g. a notation error in the working where the following line of working makes the candidate's intent clear.

MARK SCHEME NOTES

The following notes are intended to aid interpretation of mark schemes in general, but individual mark schemes may include marks awarded for specific reasons outside the scope of these notes.

Types of mark

M Method marks, awarded for a valid method applied to the problem.

A Accuracy mark, awarded for a correct answer or intermediate step correctly obtained. For accuracy marks to be given, the associated Method mark must be earned or implied.

B Mark for a correct result or statement independent of Method marks.

When a part of a question has two or more 'method' steps, the M marks are in principle independent unless the scheme specifically says otherwise; and similarly where there are several B marks allocated. The notation 'dep' is used to indicate that a particular M or B mark is dependent on an earlier mark in the scheme.

Abbreviations

awrt answers which round to
cao correct answer only
dep dependent
FT follow through after error
isw ignore subsequent working
nfww not from wrong working
oe or equivalent
rot rounded or truncated
SC Special Case
soi seen or implied

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Question	Answer	Marks	Partial marks
1	−0.8, 0.55, 2.25	**B1**	
	$x < -0.8$ $0.55 < x < 2.25$	**B2**	**B1** for either inequality correct
2(a)	$5 - (x + 2)^2$	**B2**	**B1** for $-(x + 2)^2$ or $(x + 2)^2$ or $a = 5$ and $b = 2$
2(b)	$f \leqslant 5$ or $f(x) \leqslant 5$	**B1**	**FT** *their* 5
2(c)	−2	**B1**	**FT** − *their* 2
2(d)	Complete method to find inverse function: Swaps the variables and rearranges or rearranges and swaps the variables at some point in their solution	**M1**	For **M1 FT** *their* part (a) provided it is in the form $a-(x+b)^2$
	$[g^{-1}(x) =] -2+\sqrt{5-x}$ or $\left[g^{-1}(x)=\right]\dfrac{-4+\sqrt{4^2-4[1](x-1)}}{2}$ or $\left[g^{-1}(x)=\right]\dfrac{-(-4)-\sqrt{(-4)^2-4(-1)(1-x)}}{-2}$ oe, isw	**A2**	**A1** for $[g^{-1}(x) =] -2\pm\sqrt{5-x}$ or $[g^{-1}(x) =] =\dfrac{-4\pm\sqrt{4^2-4[1](x-1)}}{2(1)}$ or $[g^{-1}(x) =] =\dfrac{-(-4)\pm\sqrt{4^2-4(-1)(1-x)}}{2(-1)}$
	[Domain:] $x \leqslant 5$	**B1**	**FT** *their* part (b) provided it is in form $f(x) \leqslant a$ where a is a constant
	[Range:] $g^{-1} \geqslant$ *their* −2	**B1**	**FT** *their* value of k in part (c)
3(a)	$4\tan^2\theta - \sec^2\theta$	**M1**	e.g. $4\dfrac{\sin^2\theta}{\cos^2\theta}-\dfrac{1}{\cos^2\theta}$
	Justified completion to given answer e.g. $4\tan^2\theta - (1 + \tan^2\theta)$ $= 3\tan^2\theta - 1$ or $3\tan^2\theta - (\sec^2\theta - \tan^2\theta)$ $= 3\tan^2\theta - 1$	**A1**	e.g. $4\dfrac{\sin^2\theta}{\cos^2\theta}-\dfrac{\sin^2\theta+\cos^2\theta}{\cos^2\theta}$ $= 4\tan^2\theta - \tan^2\theta - 1$ $= 3\tan^2\theta - 1$
3(b)	$\tan\theta = [\pm]\sqrt{\dfrac{2}{3}}$ oe	**M2**	**M1** for $\tan^2\theta = \dfrac{2}{3}$ oe
	$[\theta =]$ 39.2 or 39.23 to 39.232 $[\theta =]$ 140.8 or 140.76 to 140.77 and no extras in range	**A2**	**A1** for either one correct, ignoring extras

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Question	Answer	Marks	Partial marks
4(a)	Correct statement for area e.g. $\frac{1}{2}r^2\alpha = 9$ or $\frac{1}{2}rs = 9$	**B1**	
	Finds an equation that can be used to eliminate α or s e.g. $\alpha = \frac{18}{r^2}$ or $r\alpha = \frac{18}{r}$ or $s = \frac{18}{r}$	**M1**	
	Correct substitution and completion to given answer e.g. $P = 2r + r \times \frac{18}{r^2} = 2r + \frac{18}{r}$	**A1**	
4(b)	Correct derivative: $2 - \frac{18}{r^2}$ oe isw	**B1**	
	$2 - \frac{18}{r^2} = 0$ and solves for r	**M1**	**FT** *their* $\frac{\mathrm{d}P}{\mathrm{d}r} = 0$ providing of the form $2 + \frac{n}{r^k}$ where n and k are non-zero integers
	$r = 3$ only soi	**A1**	
	$P = 12$ only soi	**A1**	
	[2nd derivative =] $\frac{36}{r^3}$ oe **and** which is positive for $r = 3$ →minimum **or** as $r > 0$ [2nd derivative =] > 0 →minimum **or** [2nd derivative =] $\frac{36}{3^3}$ oe →minimum OR correctly finds the values of the first derivative at $3 \pm h$ where h is small → minimum	**A1**	**dep** on $r = 3$ and no other value

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Question	Answer	Marks	Partial marks
5	[When $x = 3$] $y = \frac{2}{3}$ soi	**B1**	
	Correct derivative: $\frac{x \times \frac{1}{2}(x+1)^{-\frac{1}{2}} - \sqrt{x+1}}{x^2}$ or $-x^{-2}(x+1)^{\frac{1}{2}} + x^{-1} \times \frac{1}{2}(x+1)^{-\frac{1}{2}}$	**M2**	**M1** for an attempt to differentiate using the quotient rule oe
	Gradient of tangent: $\frac{3 \times \frac{1}{2}(4)^{-\frac{1}{2}} - \sqrt{4}}{3^2}$	**M1**	**dep** on attempt at a derivative which includes $(x+1)^{-\frac{1}{2}}$
	$-\frac{5}{36}$ isw	**A1**	**dep** on having a **correct derivative**
	$y = -\frac{5}{36}x + \frac{13}{12}$ oe or $y - \frac{2}{3} = \left(-\frac{5}{36}\right)(x - 3)$ oe soi	**A1**	**FT** *their* non-zero $\frac{2}{3}$ and *their* non- zero $-\frac{5}{36}$
	$A(15, -1)$	**B2**	**B2 dep** on all previous marks awarded **B1 dep** for $x = 15$ or $y = -1$
6(a)	$\frac{2}{3}(3x+2)^{\frac{1}{2}}(+c)$ oe	**2**	**B1** for $k(3x+2)^{\frac{1}{2}}$ oe, soi where k is a constant and $k \neq 0$
6(b)	$-\frac{e^{1-2a}}{2} + \frac{1}{2}$ oe, isw	**3**	**B2** for $-\frac{1}{2}e^{(1-2x)}$ oe or **B1** for $ke^{(1-2x)}$ or $ke^{(1-2a)}$ where k is a constant, $k \neq 0$
7(a)	$x = \sqrt[3]{\frac{28}{56}}$ oe, nfww, isw	**3**	**B2** for $28x^{10}$ and $56x^{13}$ OR $[x^8](28x^2)$ and $[x^8](56x^5)$ or **B1** for $28x^{10}$ or $56x^{13}$ OR $[x^8](28x^2)$ or $[x^8](56x^5)$
7(b)(i)	$n = 10$	**2**	**B1** for the correct term in any form e.g. ${}^nC_5x^{n-5}\left(\frac{2}{x}\right)^5$ or for $n - 5 = 5$ oe, soi
7(b)(ii)	8064	**2**	**B1** for ${}^{10}C_5 \times 2^5$ oe e.g. 252×32

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Question	Answer	Marks	Partial marks
8(a)	$[x =]\ \frac{\pi}{24}, \frac{5\pi}{24}$ and no extras in range	**2**	**B1** for $[x =]\ \frac{\pi}{24}$ or $\frac{5\pi}{24}$ ignoring extras or for $4x = \frac{\pi}{6}$ **and** $\frac{5\pi}{6}$ soi
8(b)	Correct and complete plan soi e.g. $\int_{their\frac{\pi}{24}}^{their\frac{5\pi}{24}} \left(\sin(4x) - \frac{1}{2}\right) dx$ or $2\int_{their\frac{\pi}{24}}^{\frac{\pi}{8}} \left(\sin(4x) - \frac{1}{2}\right) dx$	**M1**	**FT** *their* limits in radians from part (a) or $\int_{their\frac{\pi}{24}}^{their\frac{5\pi}{24}} \sin 4x\, dx - \frac{1}{2}\left(their\frac{5\pi}{24} - their\frac{\pi}{24}\right)$
	Integrates sin 4x: $-\frac{1}{4}\cos 4x$	**B2**	**B1** for $k\cos 4x$ where $k < 0$ or $k = \frac{1}{4}$
	Substitutes exact limits in correct order: $-\frac{1}{4}\cos 4\left(\frac{5\pi}{24}\right) - \left[-\frac{1}{4}\cos 4\left(\frac{\pi}{24}\right)\right]$ oe soi	**M1**	**FT** *their* exact limits and *their* $-\frac{1}{4}\cos 4x$ providing at least **B1** awarded
	$\frac{\sqrt{3}}{4} - \frac{\pi}{12}$ or exact equivalent	**A1**	
9	$\frac{18 + 12\sqrt{10}}{2 + \sqrt{10}}$	**B1**	
	$\frac{their18 + (their12)\sqrt{10}}{2 + \sqrt{10}} \times \frac{2 - \sqrt{10}}{2 - \sqrt{10}}$	**M1**	**FT** $\frac{a + b\sqrt{10}}{2 + \sqrt{10}}$ where a and b are integers
	$\frac{36 - 18\sqrt{10} + 24\sqrt{10} - 120}{-6}$	**A1**	
	$14 - \sqrt{10}$ nfww	**A1**	**dep** on all previous marks awarded
	Alternative		
	$\frac{16 + 11\sqrt{10}}{2 + \sqrt{10}} \times \frac{2 - \sqrt{10}}{2 - \sqrt{10}}$ [+1]	**(M1)**	
	$\frac{32 - 16\sqrt{10} + 22\sqrt{20} - 110}{-6}$ [+1]	**(A1)**	
	$\frac{-78 + 6\sqrt{10}\ [-6]}{-6}$ oe	**(A1)**	e.g. $13 - \sqrt{10}$ [+1]
	$14 - \sqrt{10}$ nfww	**(A1)**	**dep** on all previous marks awarded

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Question	Answer	Marks	Partial marks
10(a)	$1.1^n * 3$	**B2**	where * is any inequality sign or = **B1** for $\frac{10\left(1.1^n-1\right)}{1.1-1} * 200$ or $\frac{10\left(1-1.1^n\right)}{1-1.1} * 200$ or for $r = 1.1$ soi
	$n\log 1.1 * \log 3$ oe or $\log_{1.1} 3$ [* n]	**M1**	**FT** $1.1^n * \textit{their}\,3$ providing **B1** has been awarded for a correct sum to n terms **and** (*their* 3) > 0
	[n =]12	**A1**	**dep** on all previous marks awarded
10(b)	$r = 2$ only nfww	**B4**	**B3** for a correct equation or equations which can be solved directly for r e.g. $[d =]\frac{[a](r-1)}{2} = \frac{[a]r(r-1)}{4}$ or $[d =]\frac{[a](r-1)}{2} = \frac{[a](r^2-1)}{6}$ or $[a](r^2 - 3r + 2)$ [= 0] oe or $\frac{2(r+1)(r-1)}{(r-1)} = 6$ or $r = \frac{4d}{2d}$ or $[r =]\frac{a+2d}{a}$ or $[r =]\frac{a+6d}{a+2d}$ **and** $a = 2d$ or **B2** for a correct equation or equations which need to be rearranged to find r e.g. $[d =]\frac{ar-a}{2} = \frac{ar^2-ar}{4}$ or $[d =]\frac{ar-a}{2} = \frac{ar^2-a}{6}$ or $[a =]\frac{6[d]}{r^2-1} = \frac{2[d]}{r-1}$ or $a(r-1) = 2d$ and $ar(r-1) = 4d$ or **either** $[r =]\frac{a+2d}{a}$ or $[r =]\frac{a+6d}{a+2d}$ **and** $4d^2 - 2ad$ [= 0] or **B1** for $ar = a + 2d$ oe and **B1** for $ar^2 = a + 6d$ oe

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Question	Answer	Marks	Partial marks
11(a)(i)	12	**2**	**B1** for $3! \times 2!$ or ${}^2P_2 \times {}^3P_3$ oe
11(a)(ii)	72	**2**	**B1** for $5! - 4! \times 2!$ oe or $6 \times 2! \times 3!$ oe
11(b)	4200	**2**	**B1** for ${}^{10}C_3 \times {}^7C_3 \left[\times {}^4C_4\right]$ oe or ${}^{10}C_4 \times {}^6C_3 \left[\times {}^3C_3\right]$ oe
12(a)	Correct derivative: $-\sin t - \cos t$	**M1**	
	$-\frac{1+\sqrt{3}}{2}$ oe or −1.37 or –1.366[02…] rot to 3 or more dp	**A1**	Mark **final answer**
12(b)	[$v = 0 \Rightarrow$] $\cos t - \sin t = 0$ $t = \frac{\pi}{4}$	**B2**	**B1** for $\cos t - \sin t = 0$
	Correct integral: $\sin t + \cos t$ (+ c)	**M1**	
	$0 = \sin 0 + \cos 0 + c$	**M1**	**FT** *their* attempt to integrate
	$s = \sin t + \cos t - 1$	**A1**	
	$\sqrt{2} - 1$ oe, isw or 0.414[21…]	**A1**	**dep** on all previous marks awarded
	Alternative		
	[$v = 0 \Rightarrow$] $\cos t - \sin t = 0$ $\rightarrow t = \frac{\pi}{4}$	**(B2)**	**B1** for $\cos t - \sin t = 0$
	Correct integral: $\sin t + \cos t$	**(M1)**	
	with limits $t = \frac{\pi}{4}$ and $t = 0$	**(A1)**	Must be in radians
	Substitutes limits into correct integral	**(M1)**	**FT** *their* $t = \frac{\pi}{4}$ from attempt at solving $v = 0$
	$\sqrt{2} - 1$ oe, isw or 0.414[21…]	**(A1)**	**dep** on all previous marks awarded
12(c)	$-s - 1$ oe isw	**B1**	

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